

Multi-Channel EEG and Machine Learning for Optimized Driver Drowsiness Detection

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Abstract—Driver fatigue detection systems are crucial to enhancing road safety and eliminating vehicle accidents stemming from distracted drivers. Fatigue drastically impairs a driver's reaction time, judgment, and attention level, which increases a driver's likelihood of being in a crash. According to research, a significant proportion of traffic accidents around the world are due to drowsy driving. We utilized EEG signals that are publicly available and were collected from four drivers wearing a NeuroSky MindWave headset. Four classifiers were trained on EEG signals after preprocessing the data (e.g., filtering) and addressing class imbalance using SMOTE. The classifier that achieved the best performance was the Random Forest classifier, which achieved 78.41% accuracy, 81% precision, and 75% recall, to discriminate between drowsy and alert states. These results provide evidence for the utility of EEG-based ML models for real-time drowsiness detection.

Keywords—Electroencephalograph (EEG), Driver drowsiness, Machine learning, Preprocessing SMOTE, Gini Impurity

I. INTRODUCTION

Drowsiness, a psycho-physiological state that lies in the middle between alertness and sleepiness, is the main reason why people drive inattentive. Due to the high frequency of traffic accidents caused by sleepiness, this problem has gotten so bad that an automated system is needed to identify and stop these negative consequences before they happen. There are many monitoring approaches for detecting the drowsiness. Physiological signs, particularly the Electroencephalogram [13] signal, which is known for its reliability and efficacy,

have been found to be the best indicators of drowsiness. The dataset utilized in this study consists of 3,735 samples and 11 features, capturing EEG-based metrics for driver drowsiness detection [1]. Key features include attention and meditation scores, ranging from 1 to 100, along with power levels across various EEG frequency bands. The target variable, classification, is binary, indicating alert (1) or drowsy (0) states. The dataset is balanced, with no missing values, ensuring its readiness for advanced modeling techniques. Attention and meditation exhibit moderate variability, with mean scores of 48.2 and 56.6, respectively. It is impossible to feed the obtained EEG signals straight into a detection technique [2]. To prepare the data for analysis, the pre-processing stage eliminates undesired artifacts like noise and blinking or muscle activity. Nonetheless, there are standard procedures including normalization, segmentation, filtering, downsampling, feature Selection [5] and artifact removal. The dataset's class imbalance issue, where cases pertaining to extreme sleepiness are typically underrepresented, was resolved using the SMOTE technique. Additionally, more instances of the minority class were synthesized using the SMOTE in order to clarify the problem of imbalance. The procedure guarantees even distribution, thereby "filtering out" bias brought on by unequal class proportions in the dataset, even though it is not a conventional filtering technique. Together, these preprocessing procedures served as a filter to clean up the dataset, removing errors and noise and getting it ready for reliable and accurate model

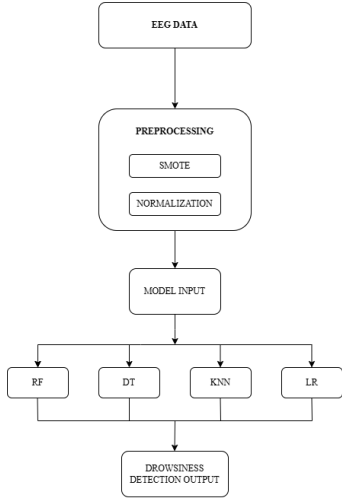


Fig. 1. Architecture for Optimized Driver Drowsiness Detection

training. By creating artificial data points for the minority class, this method enhances the model's capacity to identify infrequent occurrences like fatigue. The preprocessing technique and hyperparameter tuning [12] were relevant to the data preparation for classical machine learning models that were later used to assess the extent to which the models accurately detect driver fatigue. This research's goal is to implement a more balanced and efficient machine learning system for real-time driver fatigue detection utilizing these two approaches.

II. MATERIALS METHODS

A. Data Acquisition

The data set used in this study is comprised of 3,735 instances and 11 features, monitoring EEG signal measures of attention, meditation, and power levels of frequency bands. The target variable is binary to indicate driver states, either drowsy (0) or alert (1). The data set is suitable in the evaluation of machine learning (ML) approaches to detect driver drowsiness due to its rich signal representation and distinct labeling.

B. Data Pre-processing

Data preprocessing is an important step in developing a machine learning system to determine driver drowsiness because it prepares data to be in a balanced and ready state for model training. A heatmap was used to illustrate the initial missing values found in the dataset. For the purpose of keeping the dataset consistent and intact, any row with missing data was dropped. Besides reducing the size of the dataset, this complete-case analysis approach eliminates any biases or inaccuracies that may have been provided by partial records. The MinMaxScaler was employed to scale the feature values and render the dataset suitable for various machine learning algorithms. To avoid features with larger ranges dominating the learning process, this scaler normalized

TABLE I
SAMPLE DATASET

Attention	26	29	75	74
Meditation	34	54	87	78
Delta	960462	39145	149499	102933
Theta	277180	28225	54240	30027
Low Alpha	26575	20172	7309	10474
High Alpha	27356	39551	42355	24024
Low Beta	26575	20172	7309	10474
High Beta	13056	9933	14471	8442
Low Gamma	9126	5237	3419	3262
High Gamma	2683	4750	2269	2224
Label	0	0	1	1

all characteristics to a uniform scale ranging from 0 to 1. When working with distance-based algorithms, normalization is extremely significant because it ensures that each feature makes an equal contribution. A frequent issue in drowsiness detection datasets where non-drowsy cases often outnumber drowsy ones. SMOTE balances the dataset by interpolating between existing instances to produce synthetic examples for the minority class. Because it eliminates bias toward the majority class during training, this step is essential for enhancing the performance and generalization of models. These preprocessing steps allowed for the efficient evaluation and training of the model by removing biases and inconsistencies from the dataset.

C. Feature Normalization

A crucial preprocessing step, normalization makes that all features are scaled to a consistent range, which speeds up convergence during model training and keeps features with wider ranges from taking over the learning process. In this study, Min-Max Normalization was employed to rescale each feature x to a range of [0, 1], as defined by the equation:

$$x = \frac{\max(x) - \min(x)}{x - \min(x)} \quad (1)$$

Where:

- x original feature value,
- $\max(), \max(x)$ denotes low and high values
- x' is the normalized value.

D. Feature Selection

In the context of this study, feature extraction entails getting the dataset's natural characteristics ready for the best possible use in machine learning methods. A range of physiological or behavioral metrics (such as EEG data, reaction times, or sensor readings) that naturally contain information about driver states may be represented by the characteristics employed in this work. Since the goal was to preserve their natural representation, these raw attributes were used straight out of preprocessing in their normalized form. In this instance,

feature extraction depended on the models' innate capacity to recognize and capitalize on significant patterns in the dataset.

TABLE II
BEST HYPERPARAMETERS

MODEL	HYPERPARAMETERS
LR	['C':0.001,'penalty':'l1', 'solver':'liblinear']
K-Nearest Neighbors	['neighbors':9,'p':1, 'weights':'distance']
Decision Tree	['criterion':'gini','maxdepth':40, 'minsamples-split':5]
Random Forest	['maxfeatures':'sqrt', 'minsamples- leaf':1,'n-estimators':200]

Machine Learning Models, No explicit feature extraction was performed. The models directly utilized predefined features, such as attention, meditation, and EEG power levels. These features represent domain-specific metrics and provide the foundation for the machine learning model training. During the preprocessing and model training stages, feature extraction was carried out implicitly to prepare the data for machine learning algorithms. The many variables related to driver behavior or physiological signals in the raw dataset were scaled and balanced to increase its utility for classification tasks.

TABLE III
PREPROCESSING PARAMETERS

MODELS	PARAMETERS
KNN	StandardScaler, Normalization, FeatureSelection
LR	StandardScaler, Normalization, FeatureScaling
Random Forest (RF)	Normalization, Imputation, Fea- tureSelection,
Decision Tree (DT)	Normalization, FeatureSelection, Scaling

E. SMOTE

For addressing the imbalance between alert, drowsy samples, SMOTE was applied.

$$x_{\text{synthetic}} = x_i + \lambda \cdot (x_{\text{nn}} - x_i) \quad (2)$$

F. Model Selection

- Logistic Regression: The goal of logistic regression is to determine the ideal coefficient values so that the actual class label and the predicted probability match. These coefficients are optimized by the model using a cross-entropy function. The logistic function is given by:

$$p = \frac{1}{1 + e^{-(b_0 + b_1 X_1 + b_2 X_2 + \dots + b_n X_n)}} \quad (3)$$

where:

- Probability of positive class (awake) is denoted by p.
- K-Nearest Neighbors: KNN separates k,(distance) of the training set into the most common class in order to classify a fresh sample. The k, K-nearest neighbors are averaged for regression tasks. a non-parametric method for classifying data points by using the majority of nearest neighbors. Euclidean distance was helpful in finding the separation between two points:

$$D = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (4)$$

- Decision Tree: The tree is constructed using the feature and threshold that result in the largest impurity reduction at each node. The tree keeps growing until it hits a threshold condition. The projected class for classification is the majority class at the leaf node. a tree-based model that divides the data according to the values of its features.

Gini Impurity:

$$\text{Gini} = 1 - \sum_i p_i^2 \quad (5)$$

where:

- The probability of each class within a node is denoted by p_i .
- Random Forest: A group approach that constructs several decision trees and aggregates their forecasts to increase precision. The final forecast is calculated by averaging the estimates from each tree. The training process for each tree in the forest uses a bootstrapped sample (random sampling with replacement) of the dataset. The optimal split is obtained by considering a random selection of features for each split at a node, rather than using all available features.

G. Cross Validation

For unbalanced datasets, the StratifiedKFold makes sure that every fold keeps the same class distribution. With five folds set, the data will be divided into five parts, each of which will be used once as a test set and the other four for training. By using many data splits, cross-validation provides a precise forecast of the model's performance on data. Cuts Down on Overfitting By offering a more uniform performance assessment across various data subsets, it aids in the detection of overfitting. Optimizes the Use of Data, The fact that each data point is utilized for both testing and training makes it particularly useful for tiny datasets.

$$\text{CV Accuracy} = \frac{1}{K} \sum_{i=1}^K \frac{\text{Correct Predictions in Fold } i}{\text{Total Predictions in Fold } i} \quad (6)$$

where:

- K is count of folds.

H. Evaluation Metrics

1.Accuracy:

$$\text{Acc} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \quad (7)$$

2.Precision(P):

$$P = \frac{T_P}{T_P + F_P} \quad (8)$$

It shows the percentage of real drowsy detections out of all the drowsy samples.

3.Recall(R):

$$R = \frac{T_P}{T_P + F_N} \quad (9)$$

The percentage of accurately detected drowsy states among all actual drowsy samples is known as recall.

4.F1-Score:

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \quad (10)$$

TABLE IV
EVALUATION RESULTS OF DIFFERENT CLASSIFIERS

Model	Label(0,1)	Precision	Recall	F1-Score	Support
Logistic Regression	0	0.79	0.61	0.69	639
Logistic Regression	1	0.60	0.78	0.68	482
KNN	0	0.78	0.69	0.74	639
KNN	1	0.65	0.74	0.69	482
Decision Tree	0	0.76	0.74	0.75	639
Decision Tree	1	0.67	0.69	0.68	482
Random Forest	0	0.81	0.81	0.81	639
Random Forest	1	0.75	0.75	0.75	482

III. CLASSIFICATION

Through managing missing values, scaling features, and resolving class imbalance with SMOTE, the data preprocessing phase guarantees data ready. This phase's main objective is to produce a balanced dataset, which is essential for impartial machine learning model training. The importance of brain wave frequency bands in classifying sleepiness was highlighted by the use of mutual information scores for feature selection. To find trends pertaining to alert and sleepy states, these characteristics were taken out and examined.

For classification, four machine learning models was used, GridSearchCV was used to do hyperparameter tweaking for each model in order to determine the ideal parameters that would optimize classification accuracy. L1 and L2 penalties were used in logistic regression, whereas KNN investigated various neighbor counts and distance measures. Random Forests adjusted parameters such as n-estimators and max-features per split, while Decision Trees were tuned using depth, leaf-samples, and splitting criteria. The evaluation metrics were used to compare every model to a greater extent. In order to provide accurate measures of accuracy and loss over number of folds and to cross-validate the models for robustness, the performance and generalization capacity of the Random Forest classifier, aided by its ensemble property, was superior. Lastly, the classification results found permitted sufficient accuracy to distinguish between drowsy and alert states; this is of particular value for real-time driver monitoring systems.

A. Architecture of Machine Learning Models

Four models, i.e., Random Forest, Decision Trees, K-Nearest Neighbors (KNN), and Logistic Regression, were used to identify driver drowsiness based on EEG data. Accuracy and reliability are improved by each of the models applying certain techniques as well as hyperparameter tuning.

- Logistic Regression:

Logistic regression is a linear model which predicts the probability of a categorical outcome (for example, awake or drowsy). A linear combination of features is converted to a probability ranging from 0 to 1, via a sigmoid function. In this study, both L1 (Lasso) and L2 (Ridge) regularization were used to address multi-collinearity in the EEG feature set and reduce the likelihood of overfitting. The best regularization strength (C) parameter was assessed using GridSearchCV. For applications needing real-time predictions, Logistic Regression is computationally efficient and conceptually simple. Additionally, threshold-based decisions in safety-related use cases like driver attention gain from probabilistic results.

- K-Nearest Neighbors:

With the most votes of the closest neighbors, it is a unparametric learning algorithm to label data points. In achieving maximum accuracy, GridSearchCV was employed to tune the "k" parameter, which represents the number of neighbors considered. For determining neighbor closeness, distance metrics like the Manhattan and Euclidean distances were considered. For ensuring fair computation of distance between EEG features, the algorithm relies on feature scaling (e.g., MinMaxScaler). Due to its flexibility, KNN can handle non-linear decision boundaries, making it suitable for EEG data—which often presents complex patterns. Nevertheless, when the size of the dataset is large, particularly at inference, the

model could be computationally expensive.

- **Decision Tree:**

It is a tree-structured model that creates a hierarchy of decisions by partitioning data into subsets based on feature criteria. The maximum tree depth and the minimum samples per leaf were tuned with GridSearchCV, and the Gini impurity and entropy metrics were explored as motivations for splitting nodes. One of the principal advantages of this model is that it is interpretable, making it easy to visualize and understand the decision paths. The Decision Treemethod is effective for differentiating between beingalert and drowsy, given the method's ability to account fornon-linear associations in EEG data. In order to reduce thehabit to overfit, we pruned and modified the parametersto prevent the tree from becoming too deep.

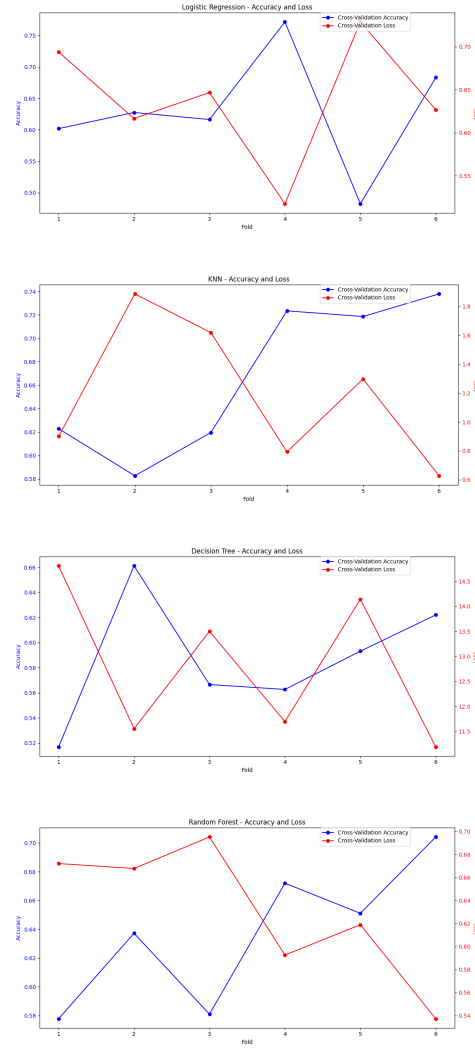
- **Random Forest:**

This model implements trees as an ensemble learning technique to improve accuracy and stability. Each tree in the forest is trained on a different subset of the data and feature attributes to promote generalization and minimize overfitting. Hyperparameters and features considered at each split are optimized using GridSearchCV. The model aggregates predictions from separate trees using majority voting to obtain good classification accuracy for the drowsiness detection challenge. Random Forest is quite good at handling the noise and unpredictability that are commonly present in EEG readings. It is one of the most reliable models in our study and outperforms the others because of its ability to capture complex feature interactions.

IV. RESULTS AND DISCUSSION

Machine learning models were used correctly to classify driver drowsiness from EEG data. With a test accuracy of 78.41% and the training accuracy of 81.92%, the Random Forest was better than others. By combining lot of decision trees, reduces the overfitting and increase in accuracy.Random Forest was a good choice for this job because it also showed balanced recall and precision. The choice Tree model's interpretability and capacity to manage intricate choice boundaries helped it perform well, with a best accuracy of 74.33% and a test accuracy of 71.99%. It produced competitive results by utilizing its distance-based classification strategy, although it was rather prone to over fitting, particularly in the absence of regularization through hyper-parameters. Its peak accuracy was 75.74%, while its test accuracy was 71.54%. Nonetheless, the model's sensitivity to the neighbors' selection and p (Minkowski distance parameter) was significant, and fine adjustment was necessary for best results.

With its straightforward methodology, logistic regression produced a test accuracy of 68.51% and a best accuracy of 68.45%. Despite performing worse than other models, it offered insightful information on the features' linear separa-



bility. Although logistic regression has trouble identifying intricate nonlinear interactions, its precision and recall measures demonstrated its capacity to distinguish between drowsy and awake or alert states.

A. Classification Report

By assessing evaluation metrics for the machine learning models offer important information about how well they predict outcomes. The Logistic Regression model, for example, demonstrated a moderate balance between accurately predicting alert and drowsy states, with slightly higher precision for identifying drivers who were not sleepy.

Its higher recall for sleepy drivers, however, demonstrated its capacity to accurately detect this crucial state. When compared to Logistic Regression, the KNN and Decision Tree models showed superior recall and precision, demonstrating their capacity to distinguish between the two groups more effectively. The Random Forest model demonstrated its superiority in correctly classifying drowsy and non-drowsy states of accuracy 78.41% while reducing misclassifications. As beneficial as these are, some models occasionally misclassified

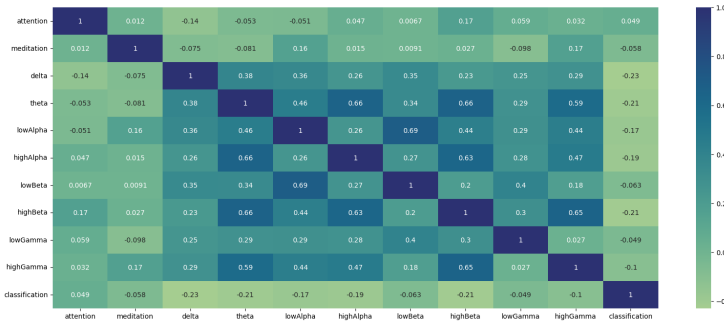


Fig. 2. Heat Map

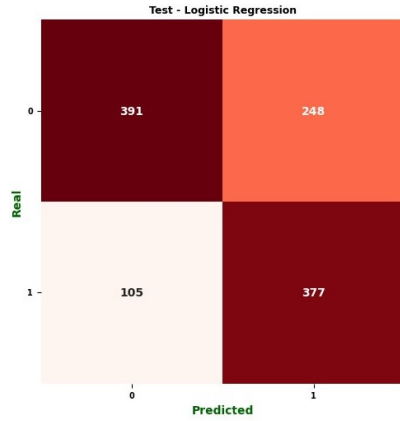


Fig. 3. Logistic Regression

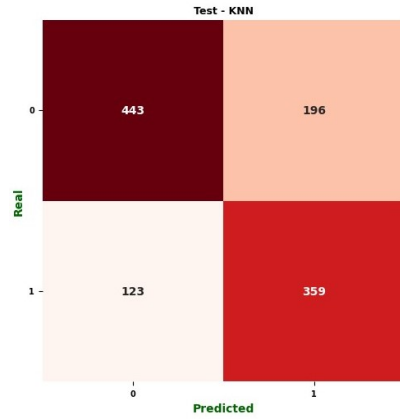


Fig. 4. K-Nearest Neighbors

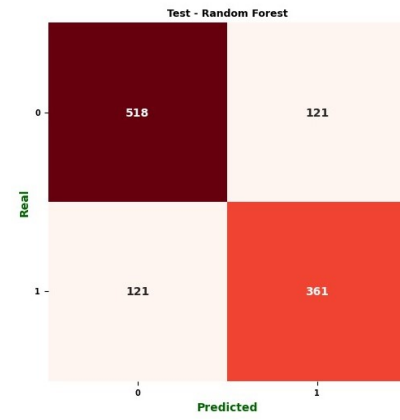


Fig. 5. Random Forest

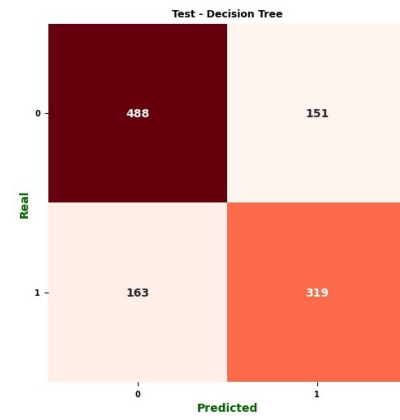


Fig. 6. Decision Tree

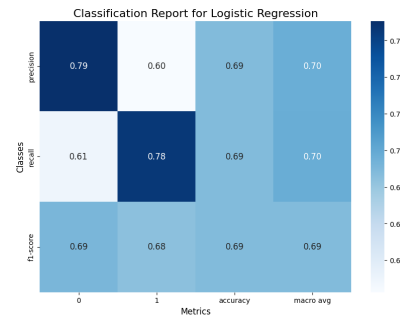


Fig. 7. Logistic Regression

sleeping drivers as awake, which is a significant issue for applications where safety is very important. The trade-off between recall and precision is critical in achieving meaningful and reliable real-world applicability of such models.

V. CONCLUSION

The study employed and evaluated a variety of machine learning models, all validated by EEG data to successfully implement driver drowsiness detection. With its metrics in

balance and more accurate outcomes than the others, the Random Forest model had the utmost advantage, with its abilities to manage difficult/nonlinear patterns present in the data set. The outcomes helped support the idea that the efficient monitoring of signs of fatigue with machine learning could improve road safety. Although, the project did show limitations as well. The primary limitation is the use of EEG data, requiring specialized equipment which may not be easily integrated into normal passenger cars. Moreover, the

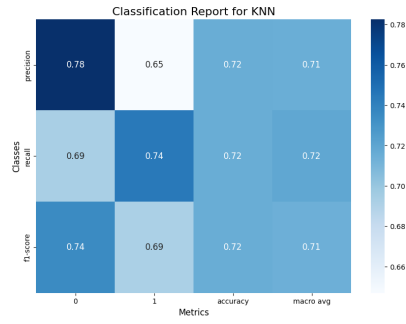


Fig. 8. KNN

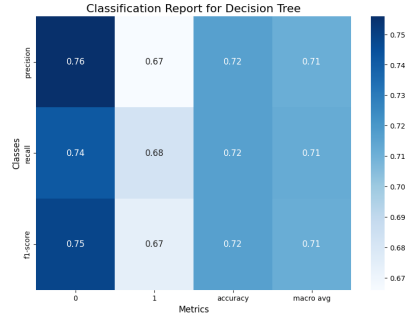


Fig. 9. Decision Tree

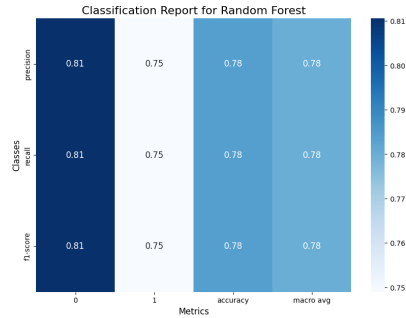


Fig. 10. Random Forest

size and variability of the dataset may inhibit the model's use in variable drivers and driving conditions.

A. Future Advancements

Future development should look at alternative sources of data that do not interfere with the driving experience, such as heart rate sensors, video monitoring, or other wearable sensors. The fusion of these alternative modalities would also allow for high variability and more accuracy and robustness of the system as a whole. Furthermore, the model's adaptability would increase as the dataset would increase and include greater variety in demographics, environmental, and driving style. Additionally, the application of deep learning methodologies such as a CNN for feature extraction and RNN associate or temporal analysis would lead to richer characterizations of driver behavior. Altogether, these future

advancements would create more pragmatic, scalable, and efficient tiredness detection systems, and ultimately contribute to safer driving experiences.

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